INCH-POUND

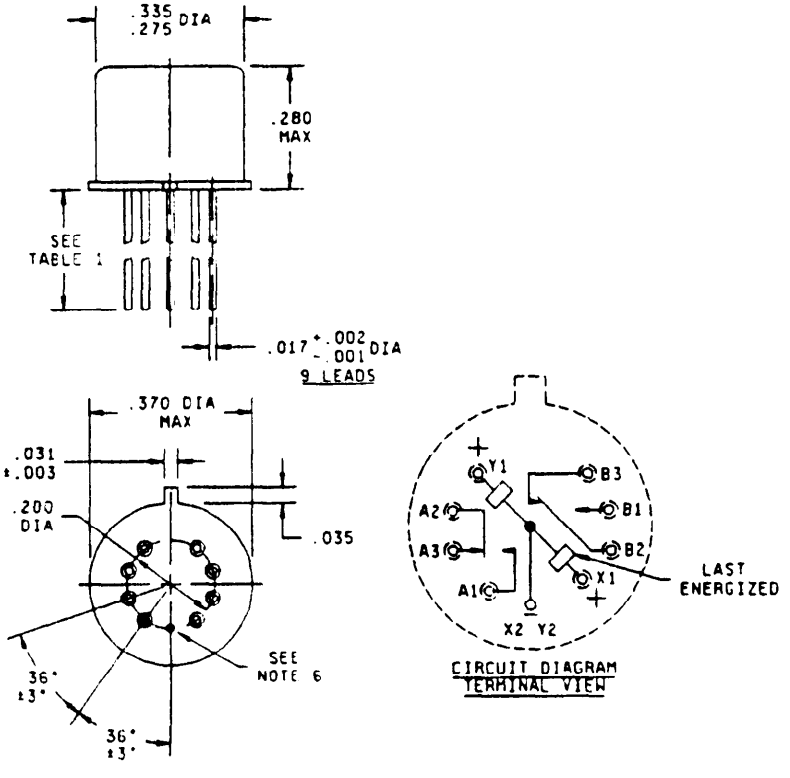
MIL-R-39016/612A
29 July 1994
SUPERSEDING
MIL-R-39016/612
20 September 1989

MILITARY SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 1.0 AMPERE (LATCHING), INCLUDING INSPECTIONS OF APPENDIXES A AND B

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.



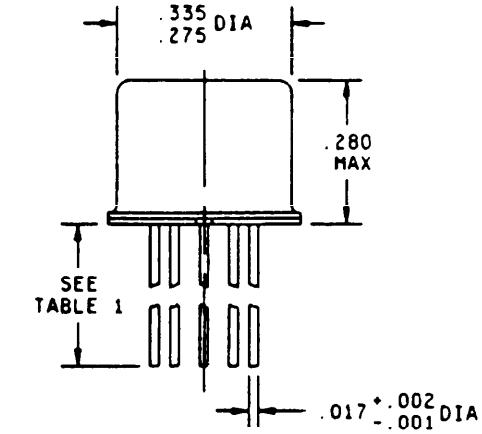
Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.200	5.08
.275	6.98
.280	7.11
.335	8.51
.370	9.40

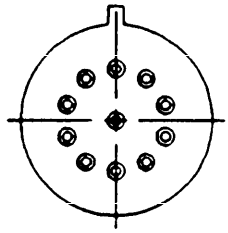
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above for reference only. Numbers do not appear on the relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. All leads shall be electrically insulated from the case, except for lead terminal -X2 -Y2, which is grounded to the case.
7. Circuit diagram shown on part is the terminal view.
8. Coil symbol optional in accordance with MIL-STD-1285.

FIGURE 1. Dimensions and configuration (round).

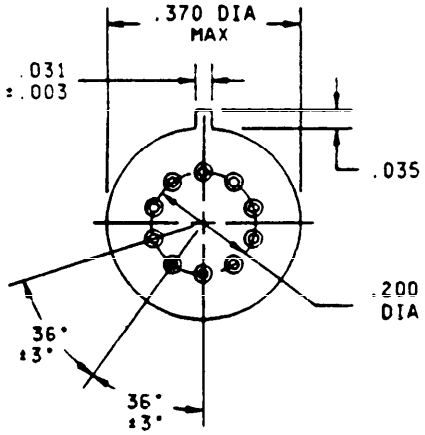
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

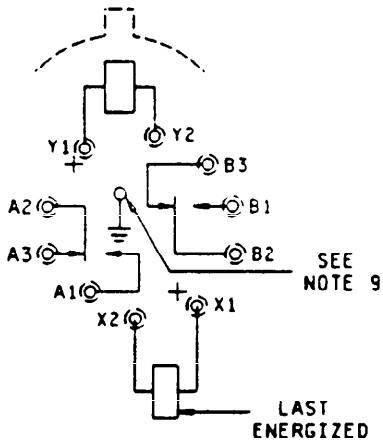
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TERMINAL VIEW
-025 THRU-030
(SEE NOTE 9)





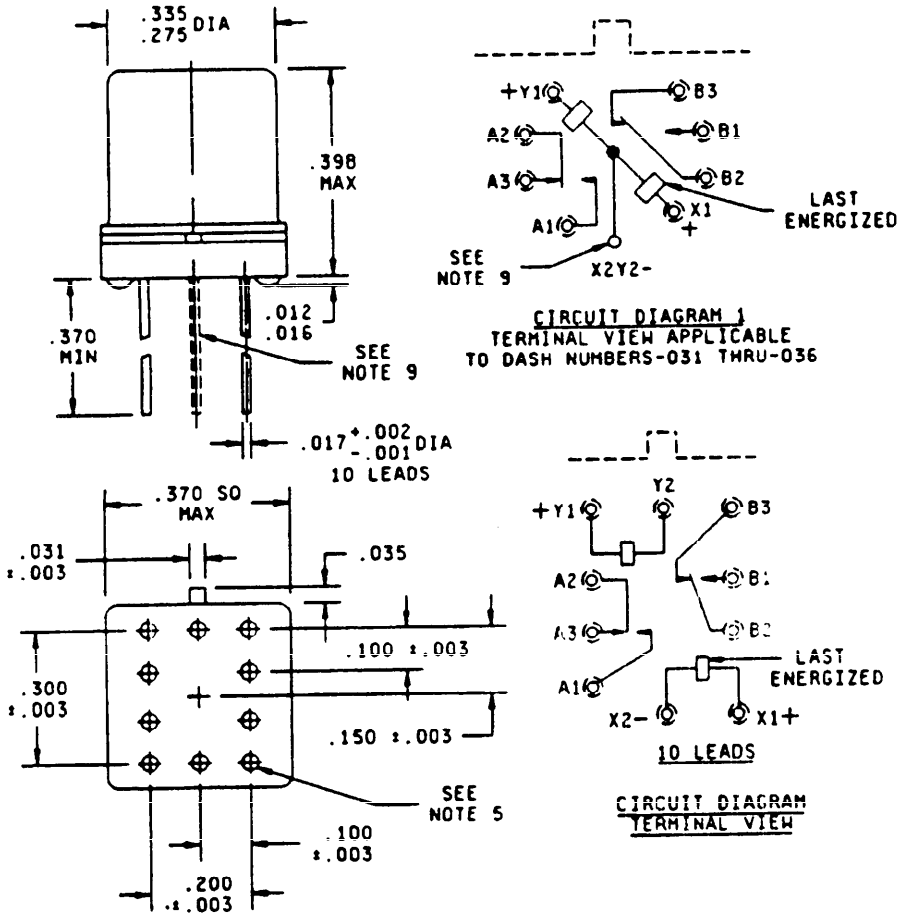
CIRCUIT DIAGRAM
TERMINAL VIEW

Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.050	1.27
.200	5.08
.280	7.11
.335	8.51
.370	9.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above for reference only. Numbers do not appear on the relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. All leads shall be electrically insulated from the case.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Circuit diagram shown on part is the terminal view.
9. M39016/612-025 through -030 shall be supplied with a case grounding pin welded to the relay header as shown.

FIGURE 2. Dimensions and configuration (square or round).

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Inches	mm
.001	0.03
.002	0.05
.003	0.08
.012	0.30
.016	0.41
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.150	3.81
.200	5.08
.300	7.62
.335	8.51
.370	9.40
.398	10.11

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader pads shall be certified to A-A-55485/5.
5. Dimensions and tolerances shown for the bottom view of the spreader pad are for the center to center location of the holes in the spreader pad.
6. Shape optional within the envelope dimension.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
9. All leads shall be electrically insulated from the case, except for lead terminal -X2 -Y2, which is grounded to the case.
10. Circuit diagram shown on part is the terminal view.
11. Terminal numbers shown above for reference only. Numbers do not appear on relay.

FIGURE 3. Dimensions and configuration relay with spreader pad attached.

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REQUIREMENTS:

Cleaning and small particle inspection: Applicable.

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:

1.0 ampere at 28 V dc.

250 mA at 115 V ac 60 Hz and 400 Hz case not grounded.

100 mA at 115 V ac 60 Hz and 400 Hz case grounded.

Inductive load: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp load: 0.10 ampere at 28 V dc.

Low level: 10 µA to 50 µA at 10 mV to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.125 ohm maximum (0.150 ohm maximum with spreader pad attached).

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.225 ohm maximum (0.250 ohm maximum with spreader pad attached).

Low level:

During life: 33 ohms maximum.

After life: 0.175 ohm maximum (0.200 ohm maximum with spreader pad attached).

Intermediate current:

During life: 1 ohm maximum.

After life: 0.225 ohm maximum (0.250 ohm maximum with spreader pad attached).

Contact bounce: 2.0 ms maximum (applicable after high level rated load life and intermediate current tests).

Contact stabilization time: 2.5 ms maximum.

Overload (high level only): Two times rated resistive current. Not applicable to ac load ratings.

Neutral screen: Applicable.

COIL DATA: See table I.

Operate time: 2.0 ms maximum over temperature range with rated coil voltage.

Release time: Not applicable.

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ELECTRICAL DATA: 1/

Insulation resistance: 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage:

Sea level:

Between case, frame or enclosure, and all contacts; both in the energized and deenergized positions	500 V rms (60 Hz).
Between case, frame or enclosure, and coils	500 V rms (60 Hz).
Between all contacts and coils	500 V rms (60 Hz).
Between open contacts in the energized and deenergized positions	500 V rms (60 Hz).
Between open contacts	500 V rms (60 Hz).
Between coils of latching relays (figure 2 and figure 3, circuit diagram)	500 V rms (60 Hz).

Altitude:

Between all terminals to case	125 V rms (60 Hz).

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): Method 204, MIL-STD-202. Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts.

Vibration (random): Method 214, MIL-STD-202, test condition IG. Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts. Applicable to qualification, group A and group C inspection testing.

Shock (specified pulse): Method 213, MIL-STD-202, test condition B (75 g's). Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with method 1041 of MIL-STD-750.

Particle Impact Noise Detection: Applicable.

PHYSICAL DATA:

Terminal strength: Method 211, MIL-STD-202.

Pull test: Test condition A, 1 pound pull.

Bend test: Test condition C, 0.5 pound pull.

Twist test: As specified in MIL-R-39016.

1/ Dielectric withstanding voltage and insulation resistance are not applicable between coil and case or from coil to coil on figure 1 relays and circuit diagram 1 relays on figure 3.

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Solderability: Applicable.

Dimensions and configuration: See figures 1, 2, and 3.

Weight: 2.84 grams (0.10 ounce) maximum, 3.09 grams (0.109 ounce) maximum with spreader pad attached.

Seal: Hermetic, adjunct sealant shall not be used.

Internal moisture: Applicable.

Minimum marking: Military Part or Identifying Number (PIN) "J" with the date code (example: J9237), circuit diagram, manufacturer's name or source code.

Outgassing: External materials (including marking ink) used in the finished relay shall be tested in accordance with ASTM-E-595. Samples to be tested shall have been processed in the same manner as that used in production of the qualification lot. Materials, when tested, shall not release greater than 1.0 percent total mass loss (TML) and 0.1 percent collected volatile condensable material (CVCM). Materials data listed in the latest revision of the NASA Reference Publication 1124 which meet these TML and CVCM requirements may be substituted in lieu of testing.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per contact load rating per relay.

Low level: 100,000 cycles per relay.

PART OR IDENTIFYING NUMBER (PIN): M39016/612- (plus dash number from table I and suffix letter designation failure rate level).

QUALIFICATION PROCEDURE: 2/

To obtain qualification to MIL-R-39016/612, a manufacturer shall:

a. Currently possess listing on QPL-39016 for all corresponding MIL-R-39016/12 dash numbers intended to be tested to MIL-R-39016/612 requirements.

b. Possess qualifying activity approved Particle Impact Noise Detection (PIND) and Cleaning and small particle inspection testing capabilities. At the discretion of the qualifying activity, approval may be granted based upon the following:

(1) An audit of the manufacturer's testing capabilities by the qualifying activity.

(2) The submission, to the qualifying activity, of procedures used in the performance of the PIND and Cleaning and small particle inspection tests.

(3) The submission, to the qualifying activity, of test data in accordance with the PIND and Cleaning and small inspection testing requirements specified in MIL-R-39016.

c. Submit to the qualifying activity, certification of intent to supply product qualified to MIL-R-39016/612.

2/ MIL-STD-790 requirements are applicable. Qualification at the initial failure rate level shall be granted based on corresponding MIL-R-39016/12 qualification. Retention of qualification and extension of qualification to lower failure rate levels shall be based on test data accumulated on MIL-R-39016/12.

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TABLE I. Dash numbers and characteristics. 1/ 2/

Dash number 3/				Coil voltage (V dc) 4/		At 25°C		At 125°C
Lead length .187 ±.010	Lead length .500 min	Figure	Spreader pad (fig. 3) 5/	Rated	Max	Coil resistance ohms ±10%	Specified pickup (latch/reset) value (voltage) (V dc)	Specified pickup (latch/reset) value (voltage) (V dc)
001	013	1	031	5.0	6.0	61	2.8	3.5
002	014		032	6.0	8.0	120	3.5	4.5
003	015		033	9.0	12.0	280	5.3	6.8
004	016		034	12.0	16.0	500	7.0	9.0
005	017		035	18.0	24.0	1,130	10.5	13.5
006	018		036	26.5	32.0	2,000	14.2	18.0
007	019	2	037	5.0	6.0	61	2.8	3.5
008	020		038	6.0	8.0	120	3.5	4.5
009	021		039	9.0	12.0	280	5.3	6.8
010	022		040	12.0	16.0	500	7.0	9.0
011	023		041	18.0	24.0	1,130	10.5	13.5
012	024		042	26.5	32.0	2,000	14.2	18.0
	025 6/			5.0	6.0	61	2.8	3.5
	026 6/			6.0	8.0	120	3.5	4.5
	027 6/			9.0	12.0	280	5.3	6.8
	028 6/			12.0	16.0	500	7.0	9.0
	029 6/			18.0	24.0	1,130	10.5	13.5
	030 6/			26.5	32.0	2,000	14.2	18.0

1/ Warning: When latching relays are installed in equipment, the latch and reset coils should not be pulsed simultaneously. Coils should not be pulsed with less than the nominal coil voltage and the pulse width should be a minimum of three times the specified operate time of the relay. If these conditions are not followed, it is possible for the relay to be in the magnetically neutral position.

2/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

3/ The suffix letter M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): M, 1.0; P, 0.1; R, 0.01. Example: 025M - - - - 030R.

4/ Caution: The use of any coil voltage less than the rated voltage will compromise the operation of the relay.

5/ Relays supplied with spreader pads (-031 through -042) shall have the pad rigidly attached.

6/ Relays are supplied with a case grounding pin welded to the header (see figure 2).

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Navy - AS, MC, OS, SH
Air Force - 19, 99
NSA - NS

Preparing activity:
DLA - ES

(Project 5945-0918-02)

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Sea Level: